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Unit – 3 : Session – 4 : SLO – 2 | Stack Applications — Infix to Postfix Conversion

■ I. Algorithm: Infix to Postfix Using a Stack

- Step 1: Scan the infix expression from left to right.
- Step 2: If the scanned symbol is an OPERAND, add it to the postfix expression.
- Step 3: If the scanned symbol is '(', PUSH it onto the stack.
- Step 4: If the scanned symbol is ')', POP and output all symbols until '(' is found; discard '('.
- Step 5: If the scanned symbol is an OPERATOR:
 - a) If the stack is empty or top is '(', push the operator.
 - b) Else, pop all operators from the stack whose precedence is GREATER THAN OR EQUAL to the scanned operator, output them, then push the scanned operator.
 - c) Note: '^' is right-associative — pop only operators of strictly higher precedence.
- Step 6: Repeat Steps 1-5 until the entire infix expression is scanned.
- Step 7: Pop and output all remaining operators from the stack.
- Step 8: EXIT

■ II. Example: $A + B * C \rightarrow ABC*+$

- Scan A $\rightarrow$ output A. Stack: empty
- Scan + $\rightarrow$ push. Stack: +
- Scan B $\rightarrow$ output B. Output: AB
- Scan * $\rightarrow$ * has higher precedence than + $\rightarrow$ push. Stack: + *
- Scan C $\rightarrow$ output C. Output: ABC
- End $\rightarrow$ pop * and +. Output: ABC*+